

AI Lead Qualifier & Outreach Automation

End-to-end n8n workflow — technical & process documentation

AI Automation & n8n Workflow Developer

n8n

OpenAI

Google Sheets

Gmail

Slack

This report documents a production-ready n8n automation that turns every inbound website lead into a scored, logged and answered conversation — without manual triage. It combines webhook intake, data validation, lightweight web scraping for context, AI-based lead scoring (BANT framework), a Google Sheets CRM, category-based email responses and an unattended daily follow-up cycle.

1. Executive Summary

Agencies, freelancers and consultants that receive leads through a web form typically lose time (and leads) to manual reading, scoring and replying. This workflow removes that bottleneck by automatically qualifying every lead the moment it arrives, replying with the right message within seconds, and following up on warm prospects who go quiet — all without a human touching the inbox.

Metric	Detail
Trigger	Webhook (POST) from any website form or CRM
Qualification engine	OpenAI GPT-4o-mini scoring 1–10 using BANT signals
Data store	Google Sheets, used as a lightweight CRM
Outbound channels	Gmail (personalised replies) + Slack (sales alerts)
Automation cycle	Real-time intake + a scheduled 9:00 AM daily follow-up job
Safeguards	Email validation, duplicate detection, private-URL blocking, prompt-injection guardrails

2. Business Value

- Zero-delay response — leads get a reply within moments of submitting the form, before interest cools.
- Consistent qualification — every lead is scored the same way, removing guesswork and rep-to-rep variance.
- Sales team only sees what matters — Slack alerts fire exclusively for Hot leads, cutting notification noise.
- Nothing falls through the cracks — the daily follow-up job automatically re-engages Hot/Warm leads that went quiet.
- No duplicate outreach — repeat submissions and already-replied leads are detected and skipped automatically.

3. Architecture Overview

The workflow is split into two independent flows that share the same Google Sheet as their source of truth.

Flow	Trigger	Purpose
Flow 1 — New Lead Intake	Webhook (instant, on form submit)	Validate, deduplicate, enrich, AI-score and respond to a new lead
Flow 2 — Daily Follow-up	Schedule (9:00 AM daily)	Re-engage Hot/Warm leads who have not replied within their follow-up window

Lead scoring uses the BANT framework (Budget, Authority, Need, Timeline). The AI model receives the lead's form data plus a scraped snippet of their company website as supporting context, and returns a structured score, category and a personalised reply opener.

Score	Category	Meaning	Follow-up timing
8–10	Hot	Clear actionable need, strong budget/buying signal, strong fit	After 1 day if no reply
5–7	Warm	Genuine interest, but budget/authority/timeline incomplete	After 3 days if no reply
1–4	Cold	Vague request, weak buying intent, or poor fit	No automated follow-up

4. Flow 1 — New Lead Capture & Qualification

Step-by-step process

- 1 Capture the lead**
A **Webhook** node listens for a POST request on /lead-capture, receiving whatever the website form or CRM sends (name, email, company, message, budget, website).
- 2 Normalize & validate**
All fields are trimmed and cleaned, the email is lower-cased and format-checked, budget defaults to “Not specified” if missing, and the company website is resolved — either from the form field or, if absent, inferred from the email domain (skipped for free providers like Gmail/Yahoo/Outlook). Local and private-network URLs (localhost, 127.x, 192.168.x, etc.) are rejected before anything is fetched.
- 3 Filter invalid emails**
Only well-formed email addresses continue; malformed submissions are stopped immediately with no further processing or credits spent.
- 4 Check for duplicates**
The Google Sheets CRM is checked by email. An existing lead triggers a Slack note and skips re-qualification, so no one is scored or contacted twice. If the lookup itself fails, a Slack alert warns the team and the lead still proceeds — the system fails safe, not silent.
- 5 Enrich with website context**
If a company website was resolved, it is fetched (8-second timeout, continues on failure) and the HTML is stripped down to roughly 2,500 characters of plain text to give the AI real business context. Leads with no website simply pass through with a placeholder.
- 6 AI qualification (GPT-4o-mini)**
The model scores the lead 1–10 on BANT signals and returns a strict JSON object: score, category, one-sentence reasoning, and a personalised reply opener referencing the lead's own message. The lead message and scraped website text are explicitly treated as untrusted data — the prompt instructs the model to never follow instructions embedded inside them.
- 7 Parse, save & route**
The AI's JSON is parsed defensively (with a safe fallback if parsing fails), the category is re-derived from the numeric score so it can never drift from the defined thresholds, the row is upserted into the Google Sheets CRM, and the lead is routed down one of three paths by category.

Category-based response

Category	Actions taken
Hot	Slack alert to the sales channel (name, company, score, reasoning) + a personalised booking email with a calendar link
Warm	A friendly email asking for budget range and timeline to complete the qualification
Cold	A polite nurture email inviting them to reach back out when ready — no follow-up scheduled

5. Flow 2 — Daily Automated Follow-up

Runs unattended every day at 9:00 AM to make sure no promising lead is forgotten.

- 1 Read every lead**
Pulls the full Google Sheets CRM into the workflow.
- 2 Filter what's due**
Keeps only Hot leads whose 1-day window has passed and Warm leads whose 3-day window has passed, excluding anyone already marked as followed up. If nothing qualifies, the run ends cleanly at a dedicated “no leads due” step instead of silently doing nothing.
- 3 Check for a reply first**
Before sending anything, Gmail is searched for a message from that lead received after their original submission. If they already replied, no follow-up email is sent — the lead is simply marked handled.
- 4 Send the right follow-up**
Hot leads get a short reminder with the booking link; Warm leads get a friendly check-in. The Google Sheets row is then updated with `followed_up = true` and a timestamp, processing one lead at a time so records stay in sync.

6. Node-by-Node Map

Stage	Key nodes
Intake & validation	Webhook - New Lead → Normalize Lead Data → IF Valid Email?
Duplicate handling	Check Duplicate Lead → IF Already Exists? → Slack Note (Duplicate) / Slack Alert (Dedup Failed)
Enrichment	IF Has Website? → Fetch Company Website → Extract Website Text / No Website Context
AI scoring	AI Qualify Lead (GPT-4o-mini) → Parse AI Output
CRM & routing	Upsert CRM (Google Sheets) → Restore Lead Context → Route by Category
Outreach	Slack Alert (Hot) / Booking Email (Hot) · Qualifying Reply (Warm) · Nurture Email (Cold)
Follow-up job	Daily 9am Trigger → Get All Leads → Filter Due Leads → Loop Over Due Leads → Check for Reply → IF Has Replied? → Send Follow-up Email / Mark Followed Up → Update Lead Record

7. Reliability & Security Design

- **Prompt-injection resistance** — the lead's own message and any scraped website text are explicitly labelled as untrusted evidence in the AI prompt, never as instructions.
- **SSRF protection** — localhost, loopback and private-network address ranges are rejected before any URL is fetched.
- **Fail-safe error handling** — a failed duplicate-check or website fetch doesn't stop the workflow; it degrades gracefully and notifies the team via Slack.
- **Deterministic categorisation** — the Hot/Warm/Cold label is recalculated from the numeric score in code, so it can never disagree with the score the AI returned.
- **Idempotent CRM writes** — leads are upserted by email, and the daily job marks each lead as followed-up individually, preventing duplicate outreach even on repeated runs.

8. Setup Checklist

- Create a Google Sheet with a “Leads” tab and the headers: name, email, company, message, budget, website, score, category, reasoning, suggested_reply_opener, timestamp, followed_up, followup_days, followup_sent_at.
 - Connect credentials: Google Sheets OAuth2, Gmail OAuth2, Slack OAuth2, OpenAI.
 - Point every Google Sheets node at the same Sheet + tab.
 - Select the Slack channel in each Slack node.
 - Replace the placeholder Calendly link in the booking and follow-up emails.
 - Update the signature and the contact URL used in the website-fetch User-Agent header.
 - Test against the webhook's Test URL, then activate and switch the form/CRM to the Production URL.
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This workflow was designed and built end-to-end in n8n as part of an AI automation portfolio, demonstrating webhook-driven intake, AI-based decisioning, CRM integration and unattended scheduled processes.